

10(a)	brighter star could be closer (to Earth)	B1
	brighter star could have a greater luminosity (in the visible wavelengths)	B1
10(b)	object with known luminosity	B1
10(c)(i)	$\frac{660.9 - 656.3}{656.3} \approx \frac{v}{3.0 \times 10^8}$ leading to $2.1 \times 10^6 \text{ m s}^{-1}$	B1
10(c)(ii)	$v = H_0 d$	C1
	$d = 2.1 \times 10^6 / 2.3 \times 10^{-18}$ $= 9.1 \times 10^{23} \text{ m}$	A1
10(c)(iii)	wavelength has increased / light is redshifted	B1
	star within galaxy is moving away / receding (from Earth)	B1
	Universe is expanding	B1